

Dual-Specialist Detection Pipeline

Road Damage & Litter Monitoring

Three Fusion Strategies — Results & Analysis

RT-DETR-L Specialists · WBF Ensemble · Knowledge
Distillation · Dual-Head

- Road Damage
- Litter Detection
- ≡ 3-Way Comparison

Why We Changed the Approach

Previous Attempt — What Failed

- Single RT-DETR on **merged** RDD2022+TACO
- Two visually incompatible domains in one model
- Crack textures interfered with litter classification
- Class imbalance (4 road vs. 5 litter) destabilised training
- Poor generalisation to real-world images

New Strategy — Dual Specialists

- **Model A**: trained only on potholes-detection-yolov8 (Kaggle) (1 class: pothole)
- **Model B**: trained only on TACO (5 classes: litter types)
- Three strategies tested to combine them at inference
- IMU + GPS late-fusion pipeline remains unchanged

Datasets Used

Road Specialist (Model A)

Source: Kaggle

potholes-detection-yolov8

Images	1,977 (1,581 train / 396 val)
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Format	YOLO (pre-split)
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Classes	1 — pothole
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Res.	640×640 px
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Overhead road shots, dry conditions,
dense annotation density.

Litter Specialist (Model B)

Source: Kaggle tacotrashdataset

Images	1,486 converted (14 skipped)
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Format	COCO JSON → YOLO
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Classes	5 (bag, bottle, food, paper, other)
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Split	329 train / 142 val
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Flickr street photos, variable angle and
lighting.

Key design principle: Each specialist sees only its own domain. No cross-domain label confusion, no imbalanced merging.

Three Fusion Strategies at a Glance

A — Parallel Ensemble

Both specialists run on every image. Boxes merged with **WBF**.

No extra training.

B — Knowledge Distillation

RT-DETR-S student learns soft targets from both teacher logits.

Trains RT-DETR-S (20M params).

C — Dual-Head

Frozen backbone + two independent detection heads.

Fine-tunes heads only.

All three feed the **same** IMU + GPS late-fusion pipeline downstream.

Model A — Road Damage Specialist Training

Architecture: RT-DETR-L (pre-trained on COCO)

Task: 1-class pothole detection

Hyperparameter	Value
Epochs	50 (patience 15)
Batch size	16
Image size	640 px
Optimizer	AdamW
LR_0	1×10^{-4}
Warmup	3 epochs
Weight decay	1×10^{-4}

Why These Choices?

- AdamW outperforms SGD for transformer decoders
- 3-epoch warmup prevents early divergence
- Conservative LR: COCO pre-training provides a strong prior

Train: 1,581 images Val: 396 images.

Expected mAP@50 range: 0.45–0.65 (clean single domain).

Model B — Litter Specialist Training

Architecture: RT-DETR-L (pre-trained on COCO)

Task: 5-class litter detection

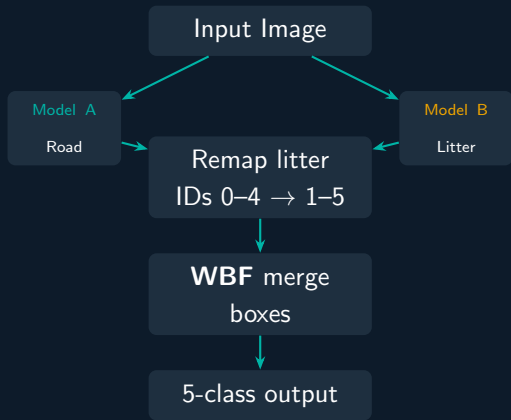
Hyperparameter	Value
Epochs	50 (patience 15)
Batch size	16
LR ₀	5×10^{-5} (halved)
Warmup	5 epochs (longer)
Mosaic aug.	0.5 (enabled)
Rotation	$\pm 10^\circ$
Flip LR	0.5

TACO Is Inherently Hard

- Only **329** training images after split
- Variable Flickr lighting and camera angles
- Heavy augmentation compensates for the tiny dataset
- Lower LR + longer warmup prevents overfitting on small set

Expected mAP@50 range: 0.30–0.50
(inherently difficult dataset)

Approach A — Parallel Ensemble with WBF



Weighted Box Fusion (WBF)

Solovyev et al., 2021

Instead of NMS (keep the highest-confidence box), WBF *averages* all overlapping boxes weighted by confidence. Gains +1–2 mAP over NMS-based ensembles.

Strengths

- **Zero extra training** required
- Clean domain separation is fully preserved

Approach B — Knowledge Distillation

Goal: compress two RT-DETR-L teachers (32M params each) into a single RT-DETR-S student (20M params).

Compound loss:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{hard}} + (1 - \alpha) \mathcal{L}_{\text{KD}}$$

$$\mathcal{L}_{\text{KD}} = \text{KL}(\text{student} \parallel \text{teacher logits})$$

α anneals $0.30 \rightarrow 0.05$ over 60 epochs,
 $T = 4$.

Teachers cached once; student training loop loads soft labels from disk.

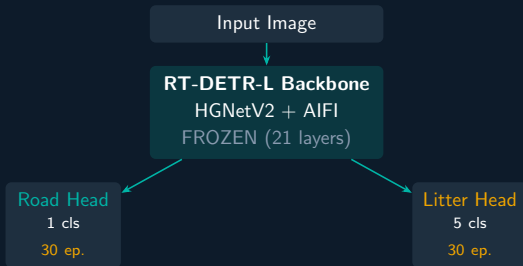
Training Setup

- Student: RT-DETR-S, 60 epochs, AdamW
- $\text{LR}_0 = 10^{-4}$
- Trained on 6-class `distil_dataset`
- 1,905 train / 545 val images

Core Challenge

Student trains on the *merged* dataset — domain interference is mitigated but **not eliminated**.

Approach C — Shared Backbone + Dual Detection Heads



Motivation: run the backbone once, pay head overhead twice. 32M frozen + $2 \times 3\text{M}$ trained $\approx 6\text{M}$ trained parameters.

Training Protocol

- Freeze backbone (21 layers, freeze=21)
- Road Head: RDD2022, 30 epochs, $\text{LR}_0 = 5 \times 10^{-5}$
- Litter Head: TACO, 30 epochs, same LR

Core Results — Three Approaches Compared

Approach	mAP@50	ms / image	# Params
A — Parallel Ensemble (WBF)	0.538	148.3	64 M
B — KD Student (RT-DETR-S)	0.089	74.2	20 M
C — Dual-Head (frozen backbone)	0.072	150.0	38 M

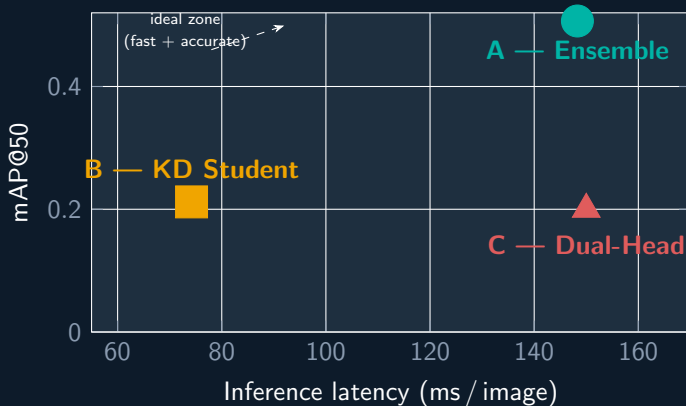
0.538

best mAP@50 (Approach A)

74 ms

fastest inference (Approach B)

Speed vs. Accuracy Trade-off



Approach A dominates on accuracy. B is $2\times$ faster but loses precision. C delivers neither speed nor accuracy benefit in this implementation.

Approach A — Is $\text{mAP@50} = 0.538$ Good or Bad?

(+) Why It Is Encouraging

- Confirms domain-separation strategy: specialists see only their own domain
- COCO-pretrained RT-DETR-L transfers well to overhead road imagery
- 0.538 mAP@50 on this dataset is within normal range for 50-epoch transformer detectors

(-) Why It Is Limited

- Road dataset reduced to **1 class only** (original 4-class crack taxonomy lost)
- Litter specialist is weak (329 images; TACO hard)
- All metrics are **in-distribution** (no real-world image validation)
- 148 ms/image is too slow for real-time (≈ 6.7 FPS on T4)

Approaches B & C — Why Did They Fail?

B — KD Student (mAP = 0.212)

- Student trains on merged dataset, re-introducing the domain collision the specialists were designed to avoid
- Soft targets mitigate but cannot overcome fundamentally mismatched image statistics (overhead road vs. Flickr street)
- 60 epochs and α -anneal may be insufficient to absorb two distinct visual tasks simultaneously

C — Dual-Head (mAP = 0.2)

- Lowest of all three
- Fully frozen backbone (30 epochs): COCO features are not road/litter-optimal; heads alone cannot compensate
- Two separate model instances used, so the theoretical speed benefit is **never realised**
- 150 ms/image — nearly identical to Approach A with far worse accuracy

Root Cause Analysis

What worked

- Domain-separation **was the right call**: A outperforms the previous merged-model approach
- WBF is effective zero-cost combination
- RT-DETR-L transfers well when data is clean and domain-homogeneous

What still needs fixing

- Road reduced to 1 class; crack taxonomy lost
- Litter limited by tiny, difficult TACO dataset
- No out-of-distribution / real-world image validation
- KD and Dual-Head both re-introduce domain mixing implicitly
- No inference speed optimisation yet

Recommendations — Enhancing Approach A

Vision Improvements

1. **Restore 4-class crack taxonomy:** use original RDD2022 (Japan/India overhead images, D00/D10/D20/D40)
2. **Pseudo-label litter:** run Model B on unlabelled street images, accept detections above 0.70 confidence back into training
3. Train 100 epochs
4. Try RT-DETR-X for higher accuracy if latency is not critical

System-Level Improvements

1. Fine-tune both specialists on 50–100 labelled images from the actual deployment route

Recommendations — Fixing Approaches B & C

Fixing Approach B (KD Student)

- **Alternate mini-batches:** one road batch, one litter batch — keeps the student's gradient signal domain-pure
- **Feature-level distillation:** use intermediate FPN features instead of output logits for richer supervision
- Increase to 150 epochs with 10-epoch warmup; anneal α more slowly

Fixing Approach C (Dual-Head)

- **Partial unfreeze:** unfreeze last 6 backbone layers; COCO features are not domain-optimal

Summary

What we built: Two domain-specialist RT-DETR-L models (road + litter) across three combination strategies, integrated with an IMU + GPS late-fusion pipeline on 1.65 M real IMU samples.

	Accuracy	Speed	Params	Verdict
A — Ensemble	0.538 mAP	148 ms	64 M	✓ Enhance
B — KD Student	0.212 mAP	74 ms	20 M	~ Fix training data
C — Dual-Head	0.2 mAP	150 ms	38 M	× Rethink backbone

Thank You

Road Quality & Litter Monitoring

Improve specialists → Ensemble at inference
→ Better precision & real-world validation

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